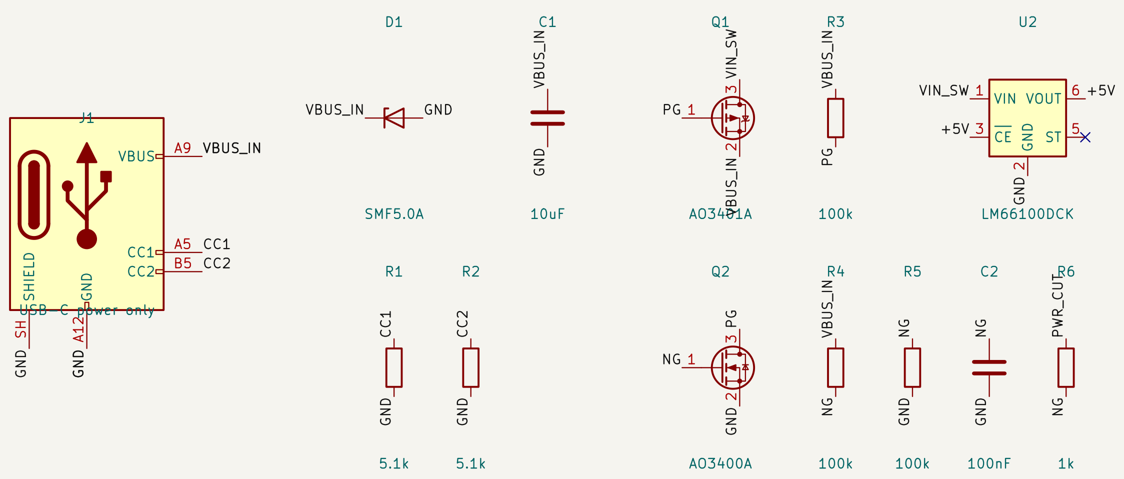
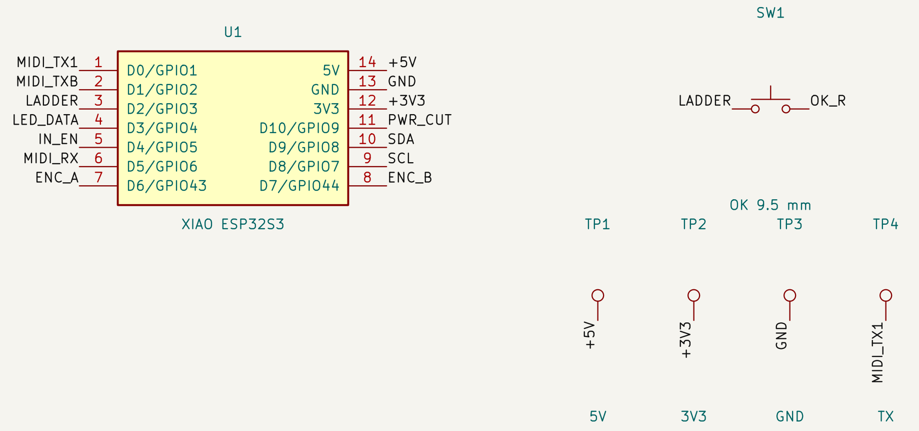
POWER IN: J1 power only (D+/D− not fitted). Q1 cut (on by default), U2 ideal diode onto the XIAO's 5V

U2 ~CE tied to VOUT: the datasheet's ideal−diode mode (blocks current back into J1). The XIAO's 5V pin IS its USB−C VBUS (no part between, Seeed v1.2 schematic): a computer on the XIAO port and a charger on J1 share +5V. Firmware pulls PWR_CUT low once a computer has enumerated it (device mode): Q2 off, Q1 off, J1 cut. At reset PWR_CUT floats: R4/R5 hold Q2 on, so a charger alone always starts it. C4 (100 uF) not fitted: host−mode hot−plug reserve, against USB's 10 uF limit for a device.

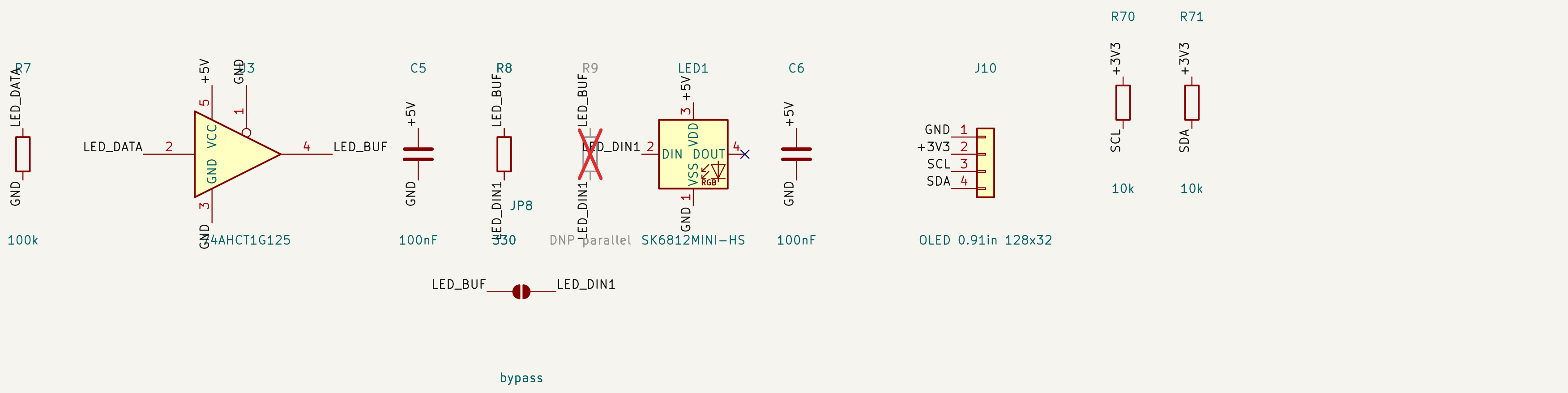

J1
VBUS_IN
CC1
CC2
USB−C power only
D1 SMF5.0A
C1 10uF
Q1 AO3401A
R3 100k
U2 LM66100DCK
VIN_SW
+5V
C3 10uF
C4 100uF DNP
R1 5.1k
R2 5.1k
Q2 AO3400A
R4 100k
R5 100k
C2 100nF
R6 1k
PWR_CUT
NG
PG

XIAO ESP32S3 (Seeed footprint, 2 x 7 headers or castellations)

D6 = GPIO43 has 499 ohm on the module (U0TXD). D2 = GPIO3 strapping, ignored by default eFuses. ENC_A = D6 = GPIO43, ENC_B = D7 = GPIO44 (U0TXD / U0RXD), IN_EN = D4 = GPIO5, MIDI_RX = D5 = GPIO6 (v5).
v5: MIDI_TX1 = D0 = GPIO1 (OUT1 alone: Hi−Z in IN mode); MIDI_TXB = D1 = GPIO2 drives OUT2, OUT3 and OUT4 together (3 loops on one pad).
LADDER = D2 = GPIO3 (ADC1): OK / BACK / PUSH on one resistor ladder. LED_DATA = D3 = GPIO4. SCL = D8 = GPIO7, SDA = D9 = GPIO8 (the OLED, I2C by the GPIO matrix). PWR_CUT = D10 = GPIO9, low = J1 cut.


U1
XIAO ESP32S3
MIDI_TX1 1 D0/GPIO1
MIDI_TXB 2 D1/GPIO2
LADDER 3 D2/GPIO3
LED_DATA 4 D3/GPIO4
IN_EN 5 D4/GPIO5
MIDI_RX 6 D5/GPIO6
ENC_A 7 D6/GPIO43
14 5V +5V
13 GND GND
12 3V3 +3V3
11 D10/GPIO9 PWR_CUT
10 D9/GPIO8 SDA
9 D8/GPIO7 SCL
8 D7/GPIO44 ENC_B
SW1
LADDER
OK_R
OK 9.5 mm
TP1 +5V 5V
TP2 +3V3 3V3
TP3 GND GND
TP4 MIDI_TX1 TX

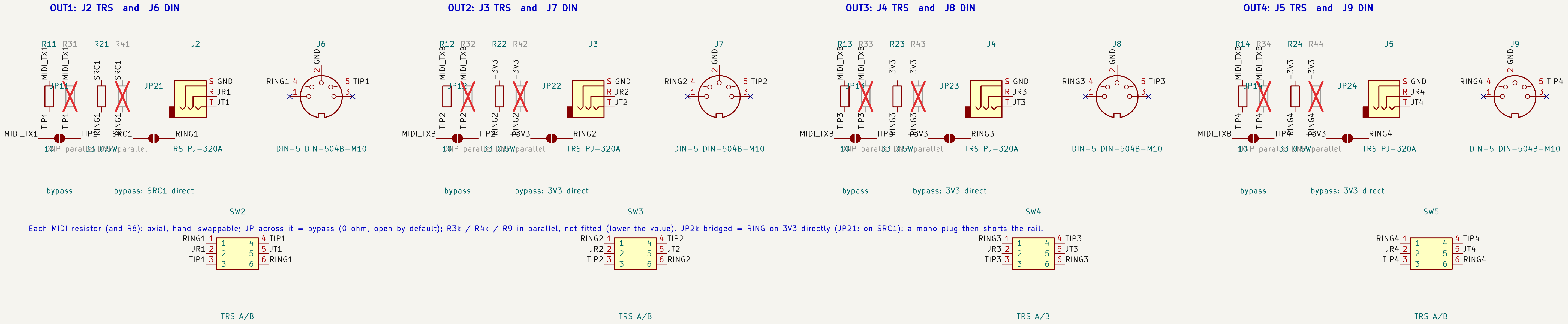
LED: one SK6812MINI−HS on +5V (top); U3 lifts the 3.3 V data to 5 V (VIH = 0.7 x VDD = 3.5 V at 5 V). SCREEN: J10, a 0.91 in 128 x 32 OLED (SSD1306, I2C 0x3C, 3V3)


GND
VBUS_IN
VIN_SW
R7 100k
U3 74AHCT1G125
LED_DATA
LED_BUF
C5 100nF
R8 330
JP8
R9 DNP parallel
LED1 SK6812MINI−HS
LED_DIN1
C6 100nF
J10 OLED 0.91in 128x32
GND
+3V3
SCL
SDA
R70 10k
R71 10k
bypass

On the board each pair is ONE footprint (J6..J9, jank:MIDI_OUT_PJ−320A_top_DIN−504B−M10_bottom: the jack's pads T, R, S carry J2..J5's nets, its switch pin SW open; the DIN's pins on the bottom). J2..J5 are BOM lines only.

MIDI OUT 1−4 (CA−033 at 3.3 V), v5: BOTH fitted on every output, the TRS (type A, top) AND the DIN−5 (bottom), in parallel; plug one at a time

TRS type A: tip = data (10 ohm to the TX pad), ring = source (33 ohm 0.5 W to 3V3), sleeve = GND. DIN−5: pin 5 = tip net, pin 4 = ring net, pin 2 = GND, pins 1 and 3 open.


OUT1: J2 TRS and J6 DIN
OUT2: J3 TRS and J7 DIN
OUT3: J4 TRS and J8 DIN
OUT4: J5 TRS and J9 DIN
R11 R31 R21 R41
R12 R32 R22 R42
R13 R33 R23 R43
R14 R34 R24 R44
JP11 JP21 JP12 JP22 JP13 JP23 JP14 JP24
10 DNP parallel 33 0.5W DNP parallel
TRS PJ−320A
DIN−5 DIN−504B−M10
bypass
bypass: SRC1 direct
bypass: 3V3 direct
MIDI_TX1
MIDI_TXB
SRC1
+3V3
TIP1 RING1 JT1 JR1
TIP2 RING2 JT2 JR2
TIP3 RING3 JT3 JR3
TIP4 RING4 JT4 JR4
SW2 SW3 SW4 SW5
TRS A/B

Each MIDI resistor (and R8): axial, hand−swappable; JP across it = bypass (0 ohm, open by default); R3k / R4k / R9 in parallel, not fitted (lower the value). JP2k bridged = RING on 3V3 directly (JP21: on SRC1): a mono plug then shorts the rail.

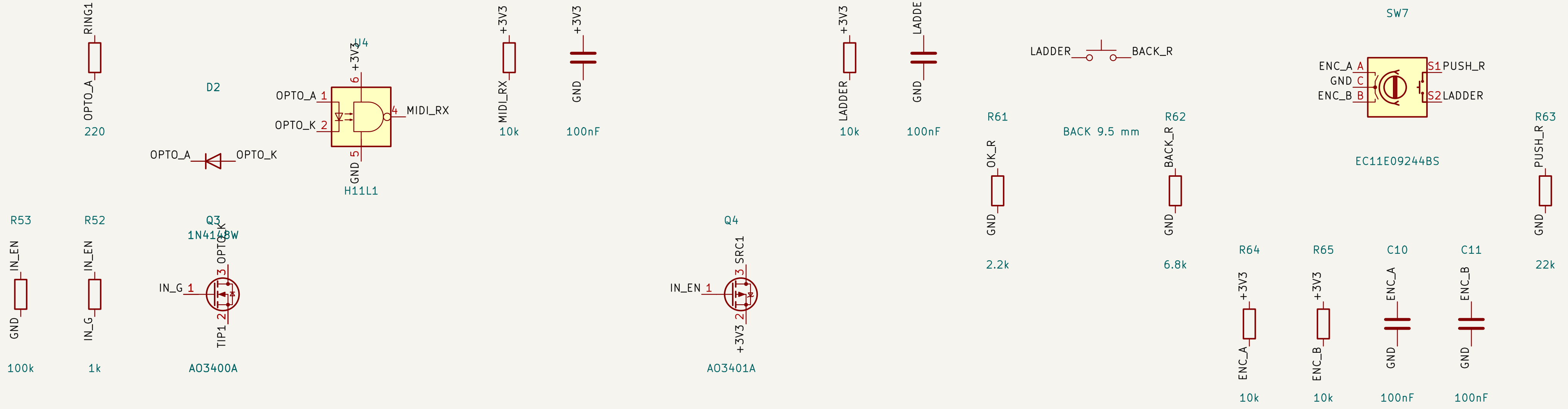
SW2..SW5: TRS type A / B per output (top side): they swap the jack's T and R between TIPk (data, sink) and RINGk (source) after the resistors; the DIN keeps pin 5 = TIPk, pin 4 = RINGk. 1−2 / 4−5 closed = type A.

OUT1 AS MIDI IN (software): TX1 to input (Hi−Z), IN_EN high (Q4 off: no source; Q3 on): MIDI_RX by UART RX

CONTROLS (top side, by the screen): one ADC ladder on GPIO3: R60 10k to 3V3; OK 2.2k, BACK 6.8k, PUSH 22k to GND

OUT mode: IN_EN low (R53 at reset): Q4 on (3V3 to the 33 ohm), Q3 off, the opto's loop open (its body diode blocks). OUT1 drives as before. D2 across the opto's LED (reverse protection). OUT1 sits on the BACK edge (long board).

Idle 3.30 V, OK 0.60 V, BACK 1.34 V, PUSH 2.27 V (one switch at a time). Encoder A/B 10k to 3V3, 100 nF to GND each.


R50 220
RING1
OPTO_A
OPTO_K
D2 1N4148W
U4 H11L1
MIDI_RX
R51 10k
C8 100nF
R53 100k
R52 1k
IN_EN
IN_G
Q3 AO3400A
Q4 AO3401A
TIP1
SRC1
R60 10k
C9 100nF
LADDER
SW6
BACK_R
BACK 9.5 mm
R61 2.2k
OK_R
R62 6.8k
SW7 EC11E09244BS
ENC_A
ENC_B
PUSH_R
R63 22k
R64 10k
R65 10k
C10 100nF
C11 100nF

generated by make_schematic.py; do not edit by hand
hardware/pcb/README.md: the power path, the pin table and every unverified item

Sheet: /
File: jank−pcb.kicad_sch
Title: jank: XIAO ESP32S3 MIDI clock stick
Size: A2 Date: 2026−11−08 **Rev: 0.1 schematic for review**
KiCad E.D.A. 10.0.8 Id: 1/1